

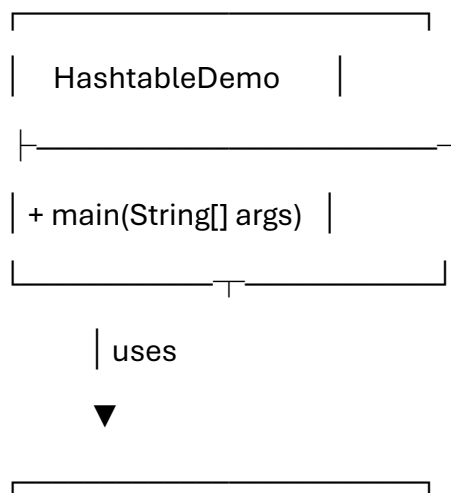
Aim

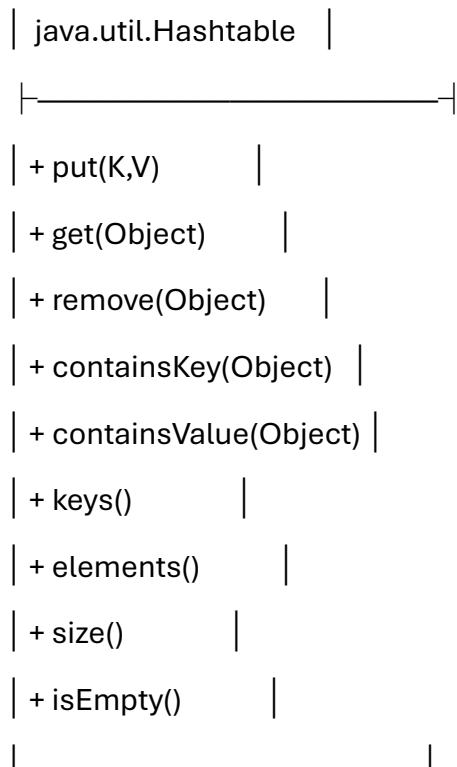
To write a Java program to demonstrate the important methods of the **Hashtable class**, such as put(), get(), remove(), containsKey(), containsValue(), keys(), elements(), size(), and isEmpty().

Algorithm

1. Start the program.
2. Import the java.util.Hashtable class.
3. Create a Hashtable object.
4. Add key-value pairs using put().
5. Display the Hashtable.
6. Retrieve a value using get().
7. Check for a key and value using containsKey() and containsValue().
8. Remove a key-value pair using remove().
9. Display all keys using keys().
10. Display all values using elements().
11. Find the number of mappings using size().
12. Check whether the Hashtable is empty using isEmpty().
13. Display the results.
14. Stop the program.

UML Diagram





Code:

```
package Observations;
```

```
import java.util.Hashtable;
```

```
import java.util.Enumeraation;
```

```
public class HashtableClass {
```

```
    public static void main(String[] args) {
```

```
        // Creating Hashtable
```

```
        Hashtable<Integer, String> table = new Hashtable<>();
```

```
        // put()
```

```
        table.put(101, "Apple");
```

```
        table.put(102, "Banana");
```

```
        table.put(103, "Mango");
```

```
        table.put(104, "Orange");
```

```
System.out.println("Hashtable: " + table);

// get()
System.out.println("Value for key 102: " + table.get(102));

// containsKey()
System.out.println("Contains key 103: "
    + table.containsKey(103));

// containsValue()
System.out.println("Contains value Mango: "
    + table.containsValue("Mango"));

// remove()
table.remove(104);
System.out.println("After removing key 104: " + table);

// keys()
System.out.print("Keys: ");
Enumeration<Integer> keys = table.keys();

while (keys.hasMoreElements()) {
    System.out.print(keys.nextElement() + " ");
}

System.out.println();
```

```

// elements()

System.out.print("Values: ");

Enumeration<String> values = table.elements();

while (values.hasMoreElements()) {
    System.out.print(values.nextElement() + " ");
}

System.out.println();

// size()

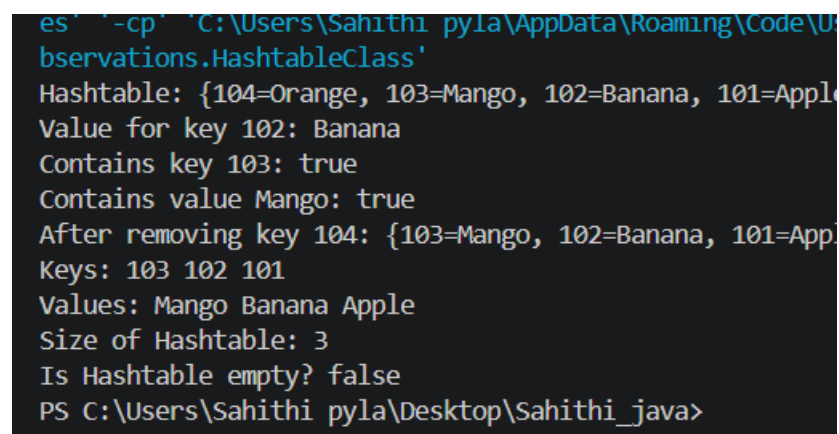
System.out.println("Size of Hashtable: " + table.size());

// isEmpty()

System.out.println("Is Hashtable empty? " + table.isEmpty());
}
}

```

Output:



```

es -cp C:\Users\Sahithi\pyla\AppData\Roaming\Code\U
bservations.HashtableClass'
Hashtable: {104=Orange, 103=Mango, 102=Banana, 101=Apple}
Value for key 102: Banana
Contains key 103: true
Contains value Mango: true
After removing key 104: {103=Mango, 102=Banana, 101=Apple}
Keys: 103 102 101
Values: Mango Banana Apple
Size of Hashtable: 3
Is Hashtable empty? false
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java>

```

Result:

Thus, the Java program successfully demonstrates the Hashtable methods put(), get(), remove(), containsKey(), containsValue(), keys(), elements(), size(), and isEmpty().

